

Inflation Experience and Housing Ownership in the U.S.

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Abstract. This paper examines the effect in inflation experiences across U. S. census regions. The study adapts an approach used in recent cross country work that links personal inflation histories to housing choices. I base my approach on the learning from experience theory developed by Melmedier and Nagel (2016). The analysis considers how higher inflation exposure could raise ownership through a desire to hedge against future price growth, but also how elevated inflation could restrict ownership through higher housing and borrowing costs. The findings aim to clarify which effect dominates across regions and to show how experiential factors interact with region level economic conditions.

1. Introduction

Every day, major financial newspapers raise questions about the future path of inflation and the interest rates expected to counterbalance it. The forward-looking nature of financial markets means that accurately predicting inflation dynamics can be both influential and highly profitable. However, I argue that there is also significant information to be gained from backward-looking data on inflation.

While the Federal Reserve can influence inflation primarily through interest rate adjustments, it cannot fully determine key household financial decisions, such as whether to purchase a home. Thus, other factors must play a significant role for people facing this decision. It may be tempting to assume a simple relationship between borrowing costs and homeownership—namely that higher inflation and rising interest rates reduce homeownership by discouraging mortgage borrowing. However, this intuitive link neither establishes causality nor captures the complexity of household decision-making. Factors such as access to credit, educational attainment, regional economic conditions, household income, and local housing prices play an important role.

Furthermore, beyond these traditional determinants, emerging research suggests that individuals’ personal experiences with inflation may also shape housing choices. In their paper “Rent or Buy? Inflation Experiences and Homeownership within and across Countries,” Ulrike Malmendier and Alexandra Steiny Wellsjo show that past inflation experiences exert a statistically significant effect on the decision to rent or buy across European countries. Their findings highlight the importance of psychological and experiential factors in long-term financial decision-making.

Motivated by their work, this paper applies a similar methodological framework to the fifty U.S. states. With a population comparable in size to that of the European Union—and substantial variation in cost of living, minimum wages, tax structures, and price dynamics—the United States provides a rich setting in which to study how inflation experiences differ across regions. Accordingly, my main research question is: Whether there is a statistically significant positive relationship between inflation experience and homeownership in the U.S.?

The central hypothesis is that higher inflation experiences may increase the likelihood of homeownership, as households seek to hedge against future price increases by purchasing

property rather than renting. Individuals who have lived through periods of high inflation have directly observed the erosion of purchasing power, while real estate assets tend to appreciate over time. However, an opposing short-run effect is also plausible: elevated inflation may reduce homeownership by increasing housing costs and mortgage rates, thereby making ownership less affordable. Given the economic diversity across U.S. states, this analysis offers the opportunity to identify where inflation experience is associated with higher homeownership—and where it may instead constrain it.

2. Literature Review

A substantial body of economic research has examined the determinants of homeownership, both within the United States and internationally. Goodman and Mayer (2018), in “Homeownership and the American Dream,” trace the evolution of U.S. homeownership rates over time and emphasize that, contrary to the widespread belief that policy alone drives homeownership, the decision to own a home carries deep financial and social significance. They argue that homeownership facilitates wealth accumulation, strengthens community ties, and provides long-term stability—highlighting why understanding its determinants is so important.

Other strands of the literature explore how institutional features and household characteristics shape housing tenure. Research on rent control policies, tax incentives, and mortgage market regulation shows that policy design can meaningfully alter the cost–benefit calculus of owning versus renting. At the household level, studies focusing on income, family size, age, education, and employment underscore the heterogeneity of housing decisions. For example, Andersen (2011), Andrews and Caldera Sánchez (2011), Bailey et al. (2019), Bracha and Jamison (2012), Collins and Choi (2010), and Drew and Herbert (2013) document how demographic and financial characteristics interact with preferences for different types of

housing—from single-family homes to apartments—and across locations and price ranges.

At a more macroeconomic level, Grevenbrock, Ludwig, and Siassi (2021) extend models of tenure choice, co-residence, and borrowing behavior to account for household decision-making under varying economic conditions. Their framework helps explain how households navigate constraints related to credit access, wealth accumulation, and borrowing costs when deciding whether to rent or own. Meanwhile, Wachter and Acolin (2018) examine the post-Great Recession recovery in homeownership and highlight its uneven nature across racial and socioeconomic groups, demonstrating that long-standing disparities continue to shape access to housing markets today. Taken together, these contributions portray homeownership as the outcome of a complex interaction among policy, macroeconomic forces, local housing market dynamics, and household characteristics.

Yet, despite this broad and growing literature, relatively little attention has been paid to the role of lived experience in shaping housing decisions. This gap is precisely where my paper contributes.

The most closely related study is Malmendier and Steiny Wellsjo (2024), “Rent or Buy? Inflation Experiences and Homeownership within and across Countries.” They provide compelling evidence that individuals’ past experiences with inflation significantly influence whether they choose to rent or buy a home. Grounded in the broader experience-effects literature (Malmendier and Nagel 2016), their analysis shows that exposure to high inflation—particularly during formative years—can lead households to favor homeownership as a hedge against future price increases.

3. Economic Theory

This section develops a theoretical framework linking inflation experience to homeownership decisions. I combine a standard utility maximization model of housing tenure choice with

the experience-based learning framework of Malmendier and Nagel(2016).

3.1. Housing Choice and Utility Maximization

The first framework relevant to this study is the classical utility maximization model, in which households choose between renting and owning in order to maximize lifetime utility. Housing provides utility both as a *consumption good*—by offering shelter, stability, and autonomy—and as an *investment*, through potential wealth accumulation from property appreciation. Homeownership is therefore an intertemporal decision that trades off current costs against future financial and non-financial benefits.

Under this framework, owning a home offers several advantages, including long-term housing security, the ability to personalize one’s living space, and the opportunity to build wealth over time as housing values rise. However, this standard model alone cannot fully explain the variation in homeownership decisions observed across individuals who face similar market conditions.

3.2. Experience-Based Learning

To capture a more realistic picture of household behavior, I build my predictions on the experience-based learning framework developed by Malmendier and Nagel. According to this theory, individuals form expectations about macroeconomic variables—such as inflation and asset returns—primarily based on their own lifetime experiences rather than on the full historical distribution of outcomes. As a result, two individuals facing identical current conditions may evaluate future risks and benefits differently depending on the inflation environments they have personally experienced.

Viewed through this lens, homeownership becomes not only an intertemporal investment decision but also one shaped by belief formation. Households weigh the current costs

of purchasing a home against expected future benefits, including equity accumulation, protection against rent volatility, and non-monetary gains such as security, autonomy, and social status.

To illustrate this intuition, consider two individuals born in year t who both face the choice to rent or own once they turn 20. Suppose one chooses to become a homeowner while the other remains a renter. Observing these choices alone does not allow us to conclude which individual made the better decision. The optimality of each choice depends on which option yields higher expected lifetime utility, given expectations about inflation, income growth, and housing prices. Assuming both individuals seek to maximize long-run utility, their decisions will depend critically on how they perceive future inflation and its effects.

3.3. Homeownership as a Hedge Against Inflation

I first model the expected lifetime utility of a household that purchases a home using a fixed-rate mortgage (according to the St. Louis Fed data around 92% of mortgages issued in the U.S. are fixed rate). The key feature of this environment is that mortgage payments are fixed in nominal terms. As inflation rises, the real burden of mortgage debt declines, making ownership more attractive in real terms.

Expected lifetime utility from owning is given by:

$$U_t^{\text{own}} = E_t \left[\sum_{k=1}^T \beta^k u \left(\frac{W_{t+k}^{\text{own}}}{1 + \pi_{t+k}} \right) \right], \quad (1)$$

where β is the discount factor and π_{t+k} denotes inflation.

The homeowner's real wealth consists of three components. First, housing wealth

appreciates over time according to:

$$H_t(1 + g_H)^k. \quad (2)$$

Second, because mortgage payments are fixed in nominal terms, their real value declines with inflation:

$$\frac{M}{1 + \pi_{t+k}}. \quad (3)$$

Finally, the household receives labor income in each period:

$$Y_{t+k}. \quad (4)$$

Combining these elements, the homeowner's real wealth is defined as:

$$W_{t+k}^{\text{own}} = H_t(1 + g_H)^k - \frac{M}{1 + \pi_{t+k}} + Y_{t+k}. \quad (5)$$

Because inflation simultaneously erodes the real value of mortgage debt and supports housing asset values, homeowners benefit in real terms from holding a leveraged housing asset during periods of rising prices. This mechanism provides a clear theoretical foundation for viewing homeownership as a hedge against inflation.

3.4. Renting Under Inflation

In contrast, I model the expected lifetime utility of a household that rents rather than buys. Unlike mortgage payments, rents typically adjust upward with inflation. As a result, renters face rising housing costs that reduce real disposable income over time.

Expected lifetime utility from renting is given by:

$$U_t^{\text{rent}} = E_t \left[\sum_{k=1}^T \beta^k u \left(\frac{Y_{t+k} - R_{t+k}}{1 + \pi_{t+k}} \right) \right]. \quad (6)$$

Rent payments evolve with inflation according to:

$$R_{t+k} = R_t(1 + \pi_{t+k}). \quad (7)$$

Because rents increase with inflation while income does not necessarily keep pace, inflation reduces renters' real purchasing power. Renters also do not benefit from housing price appreciation, as they do not own the underlying asset. Consequently, rising inflation harms renters while benefiting homeowners.

This contrast highlights the key inequality underlying the model:

$$U_t^{\text{own}} > U_t^{\text{rent}} \quad \text{when inflation rises.} \quad (8)$$

3.5. Testable Predictions

Together, this framework generates two main empirical predictions.

First, inflation experience should have a positive effect on homeownership decisions. Individuals who have lived through periods of high inflation are more likely to view homeownership as a hedge against future price increases. Empirically, this implies that the coefficient on the inflation experience variable should be positive.

Second, household income remains a central determinant of homeownership. Higher income increases access to credit and the ability to meet down payment and mortgage requirements. I expect income to have a larger quantitative effect on homeownership than inflation experience; however, income alone cannot fully explain tenure choice. Inflation

experience captures an additional belief-based channel through which households evaluate the relative attractiveness of renting versus owning.

4. Data

This study combines two primary data sources to construct measures of inflation experience and examine their relationship with homeownership outcomes. The first dataset consists of Consumer Price Index (CPI) data from the Bureau of Labor Statistics (BLS) covering the period from 1966 to 2023. The second dataset is drawn from the American Community Survey (ACS) for the years 2005 to 2023, downloaded from the IPUMS USA database.

4.1. Bureau of Labor Statistics (BLS) CPI Data

I begin by constructing annual inflation measures using the Consumer Price Index (CPI) published by the Bureau of Labor Statistics (BLS). The BLS provides CPI data at both monthly and annual frequencies; however, historical CPI series are available only at the regional and major metropolitan area levels. For the purposes of this project, I focus on CPI data aggregated at the four U.S. Census regions ‡:

- **South Region (17 states):** Alabama, Arkansas, Delaware, District of Columbia, Florida, Georgia, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, and West Virginia.
- **West Region (13 states):** Alaska, Arizona, California, Colorado, Hawaii, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming.
- **Northeast Region (9 states):** Connecticut, Maine, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont.

‡ Washington, D.C. is included as a state in the dataset; thus, there are 51 states in the regional division.

- **Midwest Region (12 states):** Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, South Dakota, and Wisconsin.

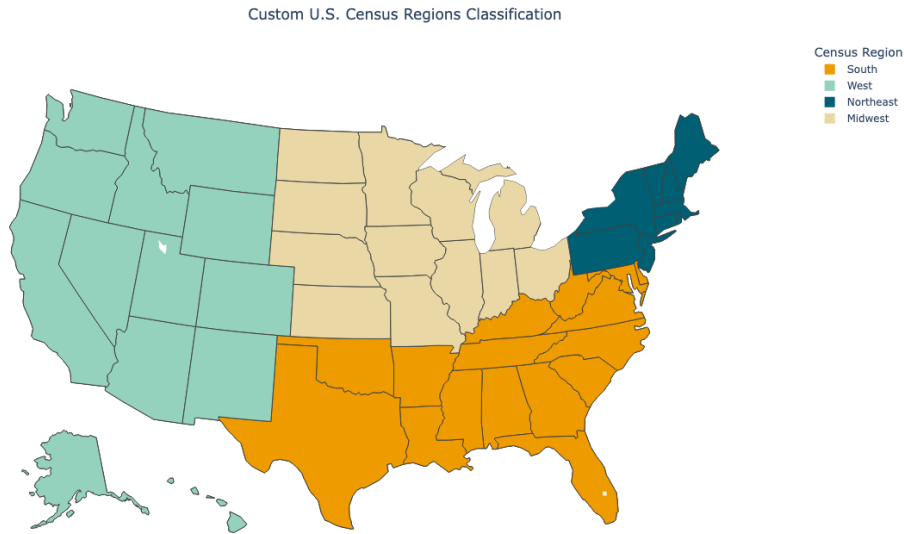


Figure 1: Illustrates the geographic aggregation of CPI data at the regional level, highlighting the limitation that inflation measures are not available at the state level for the full historical period.

To construct annual inflation rates from 1966 to 2023, I aggregate monthly CPI values to compute the annual average CPI for each region and calendar year. Annual inflation is then calculated as the percentage change in the CPI from one year to the next using the standard formula:

$$\text{Inflation Rate} = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100. \quad (9)$$

This procedure yields a region-by-year panel of annual inflation rates, which serves as the foundation for constructing individual-level inflation experience measures in the ACS sample.

Figure 2 plots annual inflation rates by Census region over time. While inflation across regions follows broadly similar national trends, meaningful differences emerge in the

timing and magnitude of inflationary episodes. As a result, individuals born in different regions—and at different points in time—are exposed to distinct inflation histories over their lifetimes.

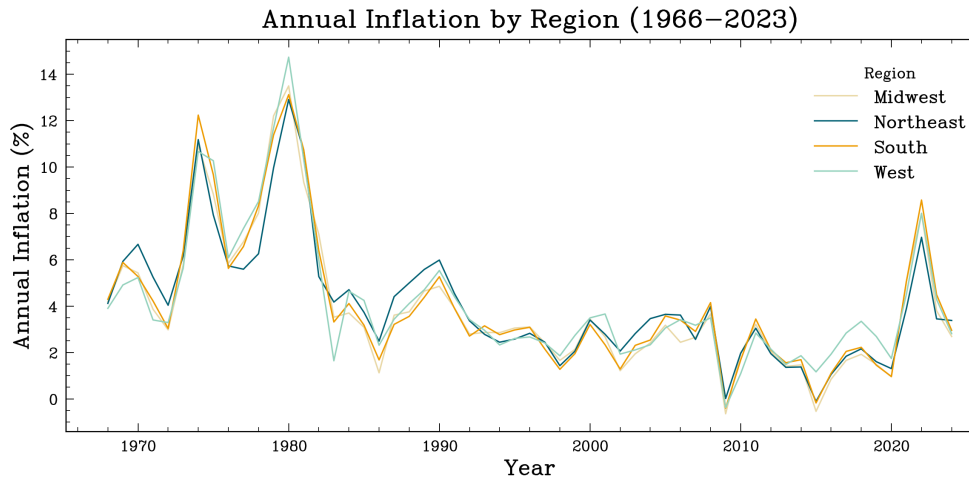


Figure 2

This cleaned and transformed CPI dataset provides the core input for constructing region-specific inflation experience measures, as follows.

4.2. American Community Survey (ACS) Data

The American Community Survey (ACS) provides annual microdata on homeownership status along with a rich set of demographic and socioeconomic characteristics - I use ACS data from 2005 to 2023. The ACS also supplies person-level survey weights, which are incorporated in all estimations to ensure representativeness of the U.S. population.

The raw ACS extract contains information on survey year, census region, state identifiers, age, marital status, educational attainment, employment status, and household income. Age and birth year are essential for constructing the inflation-experience measure, while the remaining demographic variables enter the baseline specification as controls.

4.2.1. Sample Restrictions and Data Preparation The first step in preparing the ACS data is to restrict the sample to individuals aged 20 to 40. This age range captures the population most likely to be actively making homeownership decisions, while excluding younger individuals who are unlikely to own and older individuals whose housing choices may be driven by retirement considerations.

Next, I compute each respondent's birth year by subtracting age from the survey year. Because the earliest ACS year in the sample is 2006 and the maximum allowed age is 40, the oldest possible individual in the dataset would have been born in 1966. This restriction directly informs the construction of the inflation dataset, which therefore begins in 1966 to ensure complete inflation histories for all individuals in the sample.

After generating birth years, I create a unique person identifier and harmonize regional identifiers across datasets. The ACS classifies regions using numeric codes, while the inflation data rely on region names. I therefore map each respondent's region of residence to the corresponding Census region names used in the CPI data to ensure consistency prior to merging.

4.2.2. Construction of Inflation Experience With the ACS data cleaned and regional identifiers harmonized, I construct individual-level inflation histories. For each respondent, I compile the sequence of annual inflation rates from their birth year up to their survey year, based on their region of residence.

Using this inflation history, I compute a life-cycle inflation-experience measure defined as a weighted average of annual inflation rates. Consistent with the experience-based learning framework of Malmendier and Nagel, I assign greater weight to more recent inflation observations and lower weight to more distant years, reflecting recency bias in belief formation. This procedure yields a single inflation-experience value for each individual

that summarizes their lifetime exposure to inflation.

Inflation Experience with Linear Recency Bias. Next, I want to illustrate how I used Melmedier and Nagels' learning from experience theory to calculate inflation experience. Let i index individuals and let t denote the ACS survey year. Define b_i as individual i 's birth year, and let $\pi_{r(i),s}$ denote the annual inflation rate in year s for Census region $r(i)$.

Inflation experience is defined as:

$$\text{InflExp}_{i,t} = \sum_{k=0}^{t-b_i} w_{i,t}(k) \pi_{r(i),t-k}, \quad (10)$$

where k indexes the number of years prior to the survey year and the weights satisfy $\sum_{k=0}^{t-b_i} w_{i,t}(k) = 1$.

To capture recency bias, I employ a linear decay weighting scheme in which more recent inflation realizations receive greater weight than inflation experienced earlier in life:

$$w_{i,t}(k) = \frac{(t - b_i + 1 - k)}{\sum_{j=0}^{t-b_i} (t - b_i + 1 - j)}. \quad (11)$$

Under this specification, inflation experienced in the survey year ($k = 0$) receives the highest weight, while inflation observed at birth receives the lowest weight. This linear decay structure reflects the idea that individuals place greater emphasis on more recent economic experiences when forming beliefs about future inflation, while still incorporating information from earlier periods.

4.2.3. Distribution of Inflation Experience Table 1 reports summary statistics for the inflation-experience measure across all individuals aged 20 to 40 in the ACS sample.

The median inflation experience across all regions is 2.70, while the upper tail of the

Table 1: Summary Statistics for Inflation Experience

Statistic	Value
Count	13,959,970
Mean	2.87
Std. Dev.	0.60
Min	1.78
1st Percentile	1.93
5th Percentile	2.09
10th Percentile	2.19
Median (50th Percentile)	2.70
90th Percentile	3.86
95th Percentile	4.06
99th Percentile	4.27
Max	4.35

distribution reflects substantially higher exposure to inflation, with values above 4.0 at the 95th percentile and beyond.

Shifting gears to the distribution of the inflation experience variable, the raw inflation-experience measure exhibits right skewness, driven by individuals who lived through prolonged periods of elevated inflation. While this skewness could be reduced by applying a logarithmic transformation, doing so would complicate the interpretation of regression coefficients. Given that the theoretical meaning of the inflation-experience measure is already nontrivial, I retain the original, untransformed variable for clarity in the main analysis, but keep both distributions in the following distributions to have a visual idea of what the reality of the data is for the U.S.

Furthermore, I wanted to see what the inflation experience variable looked like for each specific region in order to verify the inflation experience variable was constructed appropriately. Given that the inflation levels are felt differently in each area it would naturally follow that the inflation experience specific to each region will on average be distinct for the Northeast, Midwest, South, and West.

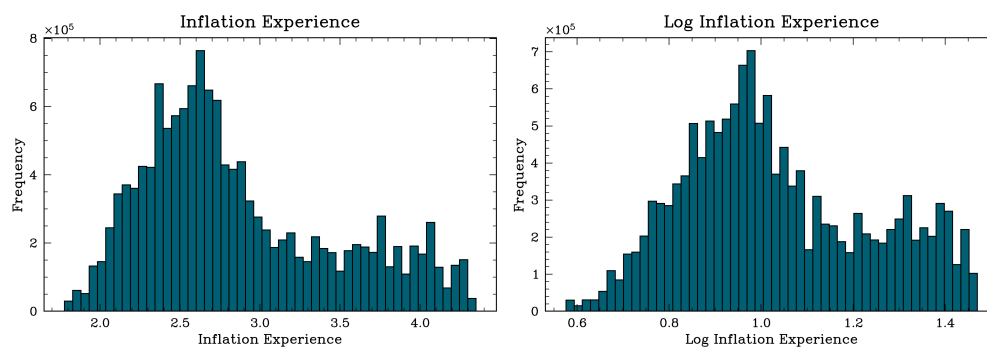


Figure 3: Distribution of inflation experience across individuals aged 20–40. The left panel shows the raw inflation-experience measure, which is right-skewed, while the right panel shows the log-transformed distribution.

Figure 4 presents the distribution of lifetime inflation experience across U.S. Census regions using box plots. While median inflation experience is broadly similar across regions, there is meaningful heterogeneity in both dispersion and upper-tail exposure. The Northeast and West exhibit slightly higher medians and wider interquartile ranges, indicating greater exposure to sustained periods of higher inflation over the life cycle.

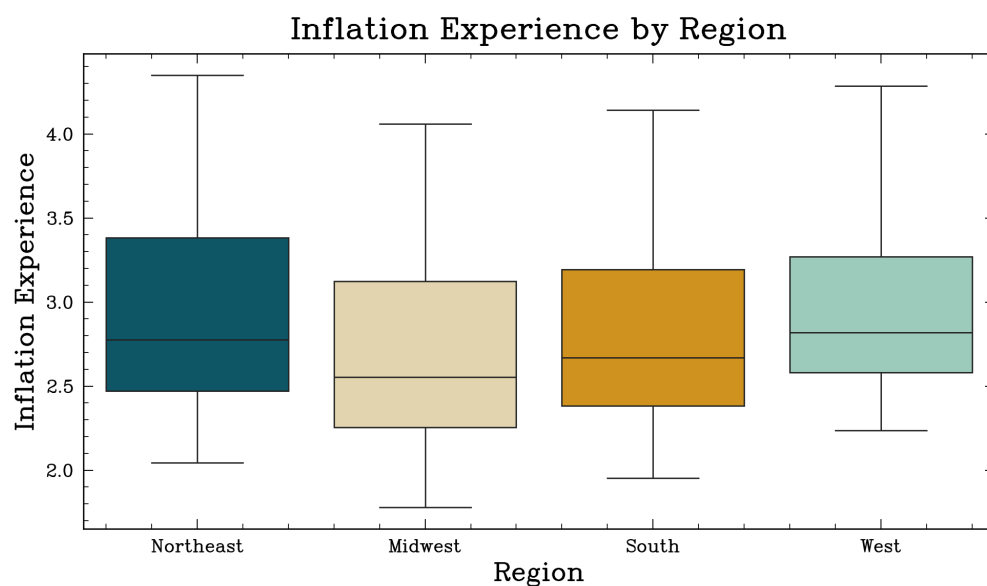


Figure 4

In contrast, the Midwest displays a lower median and a more compressed distribution,

suggesting relatively more stable inflation experiences. The South lies between these extremes, with moderate dispersion and a noticeable right tail. Across all regions, the distributions are right-skewed, reflecting a subset of individuals who experienced persistently high inflation - probably driven by individuals who experience the 1960s very high inflation levels. These regional differences motivate exploiting both within- and across-region variation in inflation experience when analyzing its relationship with housing tenure decisions.

5. Empirical Methodology

In this section, I test the predictions derived in the previous section by estimating the following econometric model:

$$\text{Homeownership}_{i,r,t} = \beta_0 + \beta_1 \text{InflationExperience}_{i,r,t} + \gamma' \mathbf{X}_{i,r,t} + \delta_t + \mu_r + \varepsilon_{i,r,t}, \quad (12)$$

where i indexes individuals, r denotes Census regions, and t denotes survey years.

The dependent variable, $\text{Homeownership}_{i,r,t}$, is an indicator equal to one if individual i residing in region r in year t reports owning their home, and zero otherwise. The key explanatory variable, $\text{InflationExperience}_{i,r,t}$, captures the individual's lifetime exposure to inflation and is constructed as a recency-weighted average of annual inflation rates observed from the individual's birth year up to year t .

The vector $\mathbf{X}_{i,r,t}$ includes a rich set of individual-level demographic and socioeconomic controls, such as age, household income, marital status, educational attainment, and employment status, which account for observable characteristics known to influence homeownership decisions.

The term δ_t represents year fixed effects, absorbing national macroeconomic conditions and common time-specific shocks—such as business cycle fluctuations or changes in monetary

policy—that affect all regions simultaneously. The term μ_r denotes region fixed effects, which control for time-invariant differences across Census regions, including persistent housing market conditions, regional institutions, and long-standing preferences for homeownership.

Finally, $\varepsilon_{i,r,t}$ is an idiosyncratic error term capturing unobserved individual-level factors affecting homeownership status. In all specifications, standard errors are robust and clustered at the Census region level to account for serial correlation within regions over time.

5.1. Baseline Regression Results

I begin by estimating a simple baseline specification that relates homeownership status to lifetime inflation experience without additional controls. Table 2 reports the results from this ordinary least squares (OLS) regression.

Table 2: OLS Regression Results: Baseline Specification

Variable	Coef.	Std. Err.	95% CI
Constant	0.3075	0.0010	[0.306, 0.309]
Inflation Experience	0.0868	0.0002	[0.086, 0.087]
Model Statistics			
Number of Observations	13,959,970		
R-squared	0.011		
Adjusted R-squared	0.011		
F-statistic	155,700		
Prob (F-statistic)	0.000		
Log-Likelihood	-9,966,000		
AIC	19,930,000		
BIC	19,930,000		
Durbin–Watson	1.256		

The coefficient on inflation experience is positive and statistically significant. Quantitatively, a one-unit increase in lifetime inflation experience is associated with an 8.7 percentage-point increase in the probability of being a homeowner. Given the large sample size, the estimate is very precisely estimated, with a small standard error and a narrow

confidence interval.

At the same time, the explanatory power of this baseline specification is limited. On one hand there must be omitted variable bias because I don't have any controls and also the R^2 is relatively low, indicating that inflation experience alone explains only a small fraction of the overall variation in homeownership outcomes. This result is expected, as homeownership decisions depend on a wide range of demographic, financial, and institutional factors beyond inflation exposure.

Motivated by both the magnitude of the baseline coefficient and the low explanatory power of the model, I next extend the specification to include a standard set of demographic and socioeconomic controls. These controls include age, household income, educational attainment, marital status, and employment status. In addition, I incorporate region fixed effects and year fixed effects to account for time-invariant differences across Census regions and common macroeconomic shocks across survey years. To ensure valid inference in the presence of within-region correlation over time, standard errors are clustered at the Census region level.

In Table 3 below, we can see that after controlling for observable characteristics and fixed effects, the coefficient on inflation experience remains positive and statistically significant but it decreased to 6.9; thus, one-unit increase in inflation experience is associated with a 6.9 percentage-point increase in the probability of being a homeowner. This attenuation suggests that part of the raw association between inflation experience and homeownership reflects correlation with demographic and socioeconomic factors. Nevertheless, the persistence of a positive and significant coefficient indicates that inflation experience continues to play an independent role in shaping homeownership decisions.

Overall, these results provide initial empirical support for the hypothesis that individuals with greater exposure to inflation are more likely to become homeowners, even after

Table 3: Panel OLS Regression Results with Region-Level Clustering

Variable	Coef.	Std. Err.	95% CI
Inflation Experience	0.0692	0.0072	[0.0552, 0.0832]
Age	0.0020	0.0006	[0.0007, 0.0032]
Household Income	-4.80×10^{-8}	8.96×10^{-10}	$[-4.97 \times 10^{-8}, -4.62 \times 10^{-8}]$
Education	0.0178	0.0013	[0.0152, 0.0203]
Marital Status	-0.0258	0.0030	[-0.0316, -0.0200]
Employment Status	-0.0080	0.0053	[-0.0183, 0.0023]
Model Statistics			
Observations			13,959,970
Entities (Regions)			4
Time Periods			18
R-squared (Overall)			0.4359
F-statistic			308,300
P-value (F-statistic)			0.0000
Log-Likelihood			-9,101,000
Included Effects: Region (Entity), Year (Time)			
Covariance Estimator: Clustered by Region			

accounting for standard determinants of housing tenure and unobserved regional and time-specific heterogeneity.

5.2. Robustness Checks

5.2.1. Education Fixed Effects As a first robustness check, I extend the empirical specification by incorporating a richer set of education fixed effects. The ACS reports educational attainment using highly detailed categories; however, many of these distinctions—such as different levels of elementary schooling or multiple categories of partial college attendance—are unlikely to be economically meaningful for analyzing homeownership decisions.

To improve interpretability and align with standard practice in applied labor and housing research, I collapse the original ACS education classifications into four broad

categories:

- Less than high school,
- High school graduate,
- Some college or associate degree, and
- Bachelor's degree or higher.

Each individual in the sample is assigned to one of these categories, and the resulting indicator variables are included as education fixed effects in the regression. This approach absorbs systematic differences in homeownership associated with educational attainment—such as variation in lifetime earnings potential, access to credit, and financial literacy—thereby reducing concerns that the estimated effect of inflation experience reflects underlying educational disparities across individuals or regions.

By conditioning on education fixed effects, identification of the inflation-experience coefficient relies on variation within education groups. The persistence of a positive and statistically significant coefficient under this specification strengthens the interpretation that inflation experience independently influences homeownership decisions, rather than merely proxying for differences in educational attainment.

The results in Table 4 indicate that inflation experience remains a powerful and statistically significant predictor of homeownership, even after absorbing detailed education fixed effects and controlling for income, age, marital status, employment, and region-year heterogeneity. A one-unit increase in lifetime inflation experience is associated with a 6.3 percentage-point increase in the probability of homeownership, holding constant both educational attainment and socioeconomic status. This finding demonstrates that the relationship is not merely an artifact of educational sorting or differences in long-term earnings capacity. Indeed, the education coefficients themselves display the expected

Table 4: Panel OLS Regression with Education Fixed Effects

Variable	Coef.	Std. Err.	t-stat	95% CI
Inflation Experience	0.0629	0.0071	8.8797	[0.0491, 0.0768]
Age	0.0029	0.0006	5.1362	[0.0018, 0.0039]
Household Income	-4.77×10^{-8}	9.57×10^{-10}	-49.871	$[-4.96 \times 10^{-8}, -4.58 \times 10^{-8}]$
Marital Status	-0.0261	0.0029	-8.9133	[-0.0319, -0.0204]
Employment Status	-0.0068	0.0049	-1.3662	[-0.0165, 0.0029]
Education Fixed Effects				
College or Higher	0.1329	0.0049	26.928	[0.1232, 0.1426]
High School Graduate	0.0866	0.0081	10.713	[0.0708, 0.1025]
Less than High School	-0.0524	0.0035	-14.802	[-0.0593, -0.0454]
Some College	0.1160	0.0097	11.920	[0.0970, 0.1351]
Model Statistics				
Observations	13,959,970			
Entities (Regions)	4			
Time Periods	18			
R-squared (Within)	0.1194			
R-squared (Between)	0.6350			
R-squared (Overall)	0.4086			
Log-Likelihood	-9,082,000			
F-statistic	2.103×10^5			
Included Effects: Region (Entity), Year (Time)				
Covariance Estimator: Robust SEs clustered at the region level				

monotonic pattern—homeownership rates are 13.3 percentage points higher for individuals with a bachelor’s degree or more, 11.6 percentage points higher for those with some college, and 8.7 percentage points higher for high school graduates, while individuals with less than a high school education are 5.2 percentage points less likely to own a home relative to the omitted category. These strong gradients confirm that education is a major channel shaping homeownership opportunities through its effects on income, credit access, and financial stability. Yet, even after accounting for these substantial differences, inflation experience continues to exert an independent and economically meaningful influence on housing tenure decisions, consistent with the experience-based learning framework.

Income Fixed Effects For the second robustness exercise, I examined whether the baseline inflation–experience coefficient was partially confounded by income heterogeneity. Household income in the ACS is highly skewed and strongly correlated with homeownership, but its influence is unlikely to be linear. To account for this potential nonlinearity, I constructed household income quintiles by ranking all individuals in the sample by reported household income and dividing the distribution into five equally sized groups. Each quintile therefore represents 20 percent of the population, ranging from the lowest–income households (Q1) to the highest–income households (Q5).

Income Quintile	Count
Q1 (Lowest)	2,792,168
Q2	2,838,594
Q3	2,747,625
Q4	2,789,989
Q5 (Highest)	2,791,594
Total	13,959,970

Replacing the continuous income measure with a full set of quintile indicators allows for flexible income fixed effects that absorb systematic differences in baseline homeownership across the income distribution, including nonlinear patterns that a simple linear control cannot capture.

Introducing these income fixed effects materially alters the estimated relationship between inflation experience and homeownership. In the baseline specification that includes demographic controls and region–year fixed effects, the coefficient on inflation experience is 0.069, indicating a strong positive association. Nonetheless, after adding the income–quintile fixed effects, we can observe in Table 5 the coefficient declines to 0.022. This reduction suggests that part of the original effect was capturing income–related variation: higher–income households are systematically more likely to own homes, and

their inflation histories may differ due to age structure, geographic mobility, or life-cycle timing. Once this income heterogeneity is fully absorbed, the remaining coefficient reflects the within-income-group behavioral effect of inflation experience on housing tenure.

Table 5: OLS Regression Results

Variable	Coef.	Std. Err.	t-stat	p-value	CI Low	CI High
Inflation Experience	0.0222	0.0072	3.0854	0.0020	0.0081	0.0363
Age	0.0051	0.0003	18.325	0.0000	0.0046	0.0057
Marital Status	-0.0271	0.0020	-13.759	0.0000	-0.0310	-0.0233
Employment Status	-0.0332	0.0051	-6.5016	0.0000	-0.0432	-0.0232
Q1 (Lowest Income)	0.2554	0.0154	16.552	0.0000	0.2252	0.2857
Q2	0.4297	0.0222	19.351	0.0000	0.3862	0.4732
Q3	0.5653	0.0238	23.713	0.0000	0.5185	0.6120
Q4	0.6698	0.0224	29.923	0.0000	0.6260	0.7137
Q5 (Highest Income)	0.5246	0.0375	13.974	0.0000	0.4510	0.5982

Crucially, although the magnitude decreases, the inflation–experience coefficient remains positive, statistically significant, and economically meaningful. Basically, one-unit increase in inflation experience is associated with a 2.2 percentage point increase in homeownership. This strengthens the interpretation that inflation experience has an independent effect on homeownership decisions, rather than serving merely as a proxy for differences in household income.

Heterogeneity Check

5.2.2. Age-Based Heterogeneity A natural question is whether the effect of inflation experience on homeownership varies across the life cycle. Younger adults and older adults face very different financial constraints, housing-market expectations, and borrowing capacity, making it plausible that the same inflation history may influence their tenure decisions in distinct ways.

To formally test for age heterogeneity, I extend the baseline regression by interacting

lifetime inflation experience with the respondent's age. This specification allows the marginal effect of inflation experience to differ at each point along the age distribution while holding constant the full set of demographic controls and region-year fixed effects. The estimated marginal effects at representative ages are reported below:

Table 6: Panel OLS Regression with Inflation–Age Interaction

Variable	Coefficient	Std. Error	95% CI
Inflation Experience	-0.1502	0.0388	[-0.2263, -0.0741]
Age	-0.0111	0.0032	[-0.0174, -0.0049]
Inflation $\times$ Age	0.0055	0.0011	[0.0034, 0.0075]
Household Income	-4.82×10^{-8}	8.53×10^{-10}	$[-4.99 \times 10^{-8}, -4.66 \times 10^{-8}]$
Education	0.0180	0.0013	[0.0154, 0.0205]
Marital Status	-0.0262	0.0029	[-0.0319, -0.0205]
Employment Status	-0.0081	0.0052	[-0.0183, 0.0022]
Model Statistics			
Observations	13,959,970		
Entities (Regions)	4		
Time Periods	18		
R-squared (Within)	0.1191		
R-squared (Between)	-1.3191		
R-squared (Overall)	-0.6907		
F-statistic	266,600		
P-value (F-statistic)	0.0000		
Log-Likelihood	-9,093,000		
Included Effects: Region (Entity), Year (Time)			
Covariance Estimator: Clustered by Region			

To interpret the interaction between inflation experience and age, I estimate age-specific marginal effects of inflation experience on the probability of homeownership. In the presence of an interaction term, the effect of inflation experience is no longer constant and varies with age. I therefore begin by estimating the following specification:

$$\begin{aligned}
 Homeownership_{i,r,t} = & \beta_0 + \beta_1 InflationExperience_{i,r,t} + \beta_2 Age_{i,r,t} \\
 & + \beta_3 (InflationExperience_{i,r,t} \times Age_{i,r,t}) + \mathbf{X}'_{i,r,t} \gamma + \mu_r + \delta_t + \varepsilon_{i,r,t}.
 \end{aligned}$$

where $\mathbf{X}_{i,r,t}$ denotes a vector of individual-level controls, μ_r are region fixed effects, and δ_t are year fixed effects. The marginal effect of inflation experience on homeownership is obtained by taking the partial derivative of homeownership with respect to inflation experience:

$$\frac{\partial \text{Homeownership}_{i,r,t}}{\partial \text{InflationExperience}_{i,r,t}} = \beta_1 + \beta_3 \text{Age}_{i,r,t}$$

Using the estimated coefficients, I evaluate this expression at ages 20 and 40 to capture how the effect of inflation experience differs across the life cycle:

$$\text{ME}(20) = \beta_1 + \beta_3 \times 20, \quad \text{ME}(40) = \beta_1 + \beta_3 \times 40.$$

The difference between these two marginal effects, $\text{ME}(40) - \text{ME}(20) = \beta_3 \times (40 - 20)$, quantifies how the influence of lifetime inflation experience on homeownership changes as individuals age, allowing for a direct comparison between younger and older cohorts. Finally displaying the following results:

- **Effect at age 20:** -0.041
- **Effect at age 40:** $+0.068$
- **Difference (40 vs. 20):** $+0.109$

These indicate substantial age heterogeneity in the behavioral response to inflation experience. For individuals around age 20, a one-unit increase in lifetime inflation experience *reduces* the probability of homeownership by approximately 4.1 percentage points. This negative effect is consistent with the notion that younger adults are more liquidity constrained and more sensitive to short-run affordability conditions: higher inflation is interpreted primarily as rising housing costs and elevated mortgage burdens, reducing the feasibility of purchasing a home.

In contrast, for individuals around age 40, the same increase in inflation experience *raises* homeownership probability by 6.8 percentage points. Older adults typically have higher incomes, stronger credit access, and accumulated savings. For them, past inflation serves as a signal that real assets such as housing provide an effective hedge against future price increases. They are therefore more likely to translate inflation experience into a proactive decision to buy a home.

The 11-percentage-point difference between ages 20 and 40 highlights a key mechanism emphasized by the experience-based learning literature: identical macroeconomic histories produce different behavioral responses depending on an individual's position in the life cycle. This pattern suggests that inflation experience influences homeownership not only directly but also through age-specific financial constraints and expectations, reinforcing the importance of modeling heterogeneity in experiential effects.

6. Results

Summary of Results

The results of this study provide strong support for both theoretical predictions. First, the coefficient on lifetime inflation experience is consistently positive and statistically significant across all specifications, confirming the prediction that individuals who have lived through higher inflation are more likely to become homeowners. Even after controlling for demographic characteristics, education groups, income quintiles, and region-year fixed effects, the effect of inflation experience remains positive and economically meaningful. This pattern aligns with the experience-based learning framework: individuals interpret past inflation as evidence of future price growth and increasingly view homeownership as a hedge that preserves real wealth. The fact that the coefficient remains robust

under alternative specifications reinforces the conclusion that inflation experience plays an independent behavioral role in shaping housing tenure decisions.

Second, the results confirm that household income is a major determinant of homeownership, with a substantially larger structural effect than inflation experience. The income quintile fixed effects reveal steep and monotonic gradients: higher-income households are dramatically more likely to own homes, reflecting differences in credit access, down-payment capacity, and financial stability. However, income alone does not fully explain homeownership decisions. When income heterogeneity is absorbed through quintile fixed effects, the inflation-experience coefficient decreases in magnitude but remains positive and significant. This demonstrates that inflation experience affects homeownership *within* income groups, not merely across them. In other words, even among households with similar income levels, those with higher inflation experience are more likely to buy a home.

Taken together, these findings suggest that while income is the dominant economic constraint governing access to homeownership, inflation experience influences whether—and when—individuals choose to transition from renting to owning. Homeownership is therefore shaped by both financial capacity and experiential expectations, indicating that behavioral responses to macroeconomic history play a meaningful role alongside traditional economic fundamentals.

6.1. Areas for Potential Development

This study extends the framework of Malmendier and Steiny Wellsjo by applying their methodology to the United States, but several limitations point to important opportunities for future work. The most significant constraint is data availability: because historical inflation data from the Bureau of Labor Statistics is published only at the Census region level, the analysis cannot exploit the substantial economic heterogeneity that exists across

states and metropolitan areas. Access to state-level or city-level historical inflation series would allow for much finer identification of how local price dynamics shape individual housing decisions. Given that housing markets are inherently local, such granularity could materially refine the estimates and reveal spatial patterns not detectable at the regional level.

A second direction for development involves completing additional robustness checks. While this paper incorporates education and income fixed effects, further analysis using alternative behavioral specifications—such as scaling the inflation-experience variable by its standard deviation—would help quantify the effect size in more intuitive units and test whether the estimated relationship is sensitive to different normalizations of the underlying measure. Due to computational limitations in the current environment, the standard-deviation scaling could not be implemented, but it remains a natural extension.

Beyond robustness, several broader empirical questions emerge from the findings and could form the basis of future work:

- **Region Dynamics:** Does living in the same region or state influence the age you chose to buy a home? For instance people in the Northeast and West had on average the higher inflation experience and based on our results they should want to buy a home as soon as possible to hedge against inflation.
- **Expectation Formation:** Does inflation experience shape individuals' expectations about future housing price growth or mortgage rates? Merging ACS microdata with survey data from the Michigan Survey of Consumers or the New York Fed's Survey of Consumer Expectations could shed light on the psychological channel behind the observed effects.
- **Credit Access and Constraints:** Does inflation experience affect the likelihood of applying for or being approved for a mortgage? Linking ACS records to HMDA

mortgage application data would allow researchers to disentangle behavioral demand from financial constraints.

Overall, while the present analysis demonstrates that inflation experience plays a meaningful and robust role in homeownership decisions, extending the empirical framework along these dimensions would deepen our understanding of how macroeconomic history shapes household behavior in real estate markets.

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